APPENDIX A: DERIVATION OF MULTI-BS LEAST-SQUARES LOCALIZATION

After obtaining the angle information of potential Eves through the VA-MUSIC method at a single BS, in order to obtain the precise physical coordinates of the target in three-dimensional space, we introduce B collaborative BSs for joint localization. The known position coordinates of the b-th BS in the three-dimensional Cartesian coordinate system is $\mathbf{w}_b = [\mathbf{x}_b, \mathbf{y}_b, 0]^T$, and the true three-dimensional coordinates of the e-th potential Eve are unknown $\mathbf{w}_e = [\mathbf{x}_e[n], \mathbf{y}_e[n], \mathbf{z}_e[n]]^T$. According to the estimation results of the VA-MUSIC method earlier, the zenith and azimuth angles of the Eve e estimated by the b-th BS are denoted as $(\hat{\theta}_{b,e}, \hat{\varphi}_{b,e})$. Based on the conversion relationship from spherical coordinates to Cartesian coordinates, the unit direction vector $\chi_{b,e}$ pointing from the b-th BS to Eve e can be expressed as:

$$\chi_{b,e} = \begin{bmatrix} \sin\hat{\theta}_{b,e}\cos\hat{\varphi}_{b,e} \\ \sin\hat{\theta}_{b,e}\sin\hat{\varphi}_{b,e} \\ \cos\hat{\theta}_{b,e} \end{bmatrix} \tag{85}$$

where the direction vector satisfies the unit length, that is, $\|\chi_{b,e}\|^2 = 1$.

In an ideal noise free environment, all rays originating from the BS position $\mathbf{w}_b$ and extending along the direction vector $\chi_{b,e}$ should precisely intersect at the true position $\mathbf{w}_e$ of the Eve. However, due to measurement and estimation errors, these rays are usually non planar straight lines in three-dimensional space and cannot intersect perfectly. Therefore, we take the square of the orthogonal distance from the target position $\mathbf{w}_e$ to the line of sight ray of the b-th BS as the error measure. Define the orthogonal projection matrix $\mathbf{P}_{b,e} \in \mathbb{R}^{3\times3}$ perpendicular to the direction vector $\chi_{b,e}$ as:

$$\mathbf{P}_{b,e} = \mathbf{I}_3 - \chi_{b,e}\chi_{b,e}^T \tag{86}$$

where $\mathbf{I}_3$ is the identity matrix of 3×3. Due to the fact that the projection matrix is idempotent (i.e. $\mathbf{P}_{b,e}^T\mathbf{P}_{b,e} = \mathbf{P}_{b,e}$, The squared Euclidean distance from the unknown target point $\mathbf{w}_e$ to the observation ray at the b-th BS can be written as:

$$\epsilon_b^2 = \|\mathbf{P}_{b,e}(\mathbf{w}_e - \mathbf{w}_b)\|^2 = (\mathbf{w}_e - \mathbf{w}_b)^T\mathbf{P}_{b,e}(\mathbf{w}_e - \mathbf{w}_b) \tag{87}$$

In order to estimate the optimal three-dimensional coordinates of Eve e, we construct a Least Squares (LS) optimization problem with the objective of minimizing the sum of squared distances from the target point to all B BS observation rays. Objective function $J(\mathbf{w}_e)$ is:

$$J(\mathbf{w}_e) = \sum_{b=1}^{B}(\mathbf{w}_e - \mathbf{w}_b)^T\mathbf{P}_{b,e}(\mathbf{w}_e - \mathbf{w}_b) \tag{88}$$

The corresponding optimization problem can be expressed as:

$$\hat{\mathbf{w}}_e = \arg\min_{\mathbf{w}_e} J(\mathbf{w}_e) \tag{89}$$

To solve the above optimization problem, we take partial derivatives of the objective function $J(\mathbf{w}_e)$ with respect to the target position vector $\mathbf{w}_e$. Using the principles of matrix calculus, the partial derivative is:

$$\frac{\partial J(\mathbf{w}_e)}{\partial \mathbf{w}_e} = 2\sum_{b=1}^{B} \mathbf{P}_{b,e}(\mathbf{w}_e - \mathbf{w}_b) \tag{90}$$

Make the partial derivative equal to the zero vector to find the global optimal solution:

$$\sum_{b=1}^{B} \mathbf{P}_{b,e}\mathbf{w}_e - \sum_{b=1}^{B} \mathbf{P}_{b,e}\mathbf{w}_b = \mathbf{0} \tag{91}$$

extracting the term containing the unknown variable $\mathbf{w}_e$ from the above equation yields a system of linear equations:

$$\left(\sum_{b=1}^{B} \mathbf{P}_{b,e}\right)\mathbf{w}_e = \sum_{b=1}^{B} \mathbf{P}_{b,e}\mathbf{w}_b \tag{92}$$

To simplify the expression, let the system matrix $\mathbf{A}_e \in \mathbb{R}^{3\times 3}$ and aggregate vector $\mathbf{c}_e \in \mathbb{R}^{3\times 1}$ be:

$$\mathbf{A}_e = \sum_{b=1}^{B} (\mathbf{I}_3 - \chi_{b,e}\chi_{b,e}^T) \tag{93}$$

$$\mathbf{c}_e = \sum_{b=1}^{B} (\mathbf{I}_3 - \chi_{b,e}\chi_{b,e}^T)\mathbf{w}_b \tag{94}$$

Finally, the estimated value of the least squares closed form solution for the three-dimensional position of Eve e is:

$$\hat{\mathbf{w}}_e = \mathbf{A}_e^{-1}\mathbf{c}_e = \left(\sum_{b=1}^{B} \mathbf{I}_3 - \chi_{b,e}\chi_{b,e}^T\right)^{-1}\left(\sum_{b=1}^{B} (\mathbf{I}_3 - \chi_{b,e}\chi_{b,e}^T)\mathbf{w}_b\right) \tag{95}$$